AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

1.Create an EC2 Windows Instance in AWS

Windows Server

Aravindraj.G

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

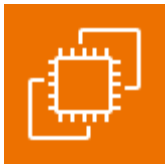
EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Windows Instances:

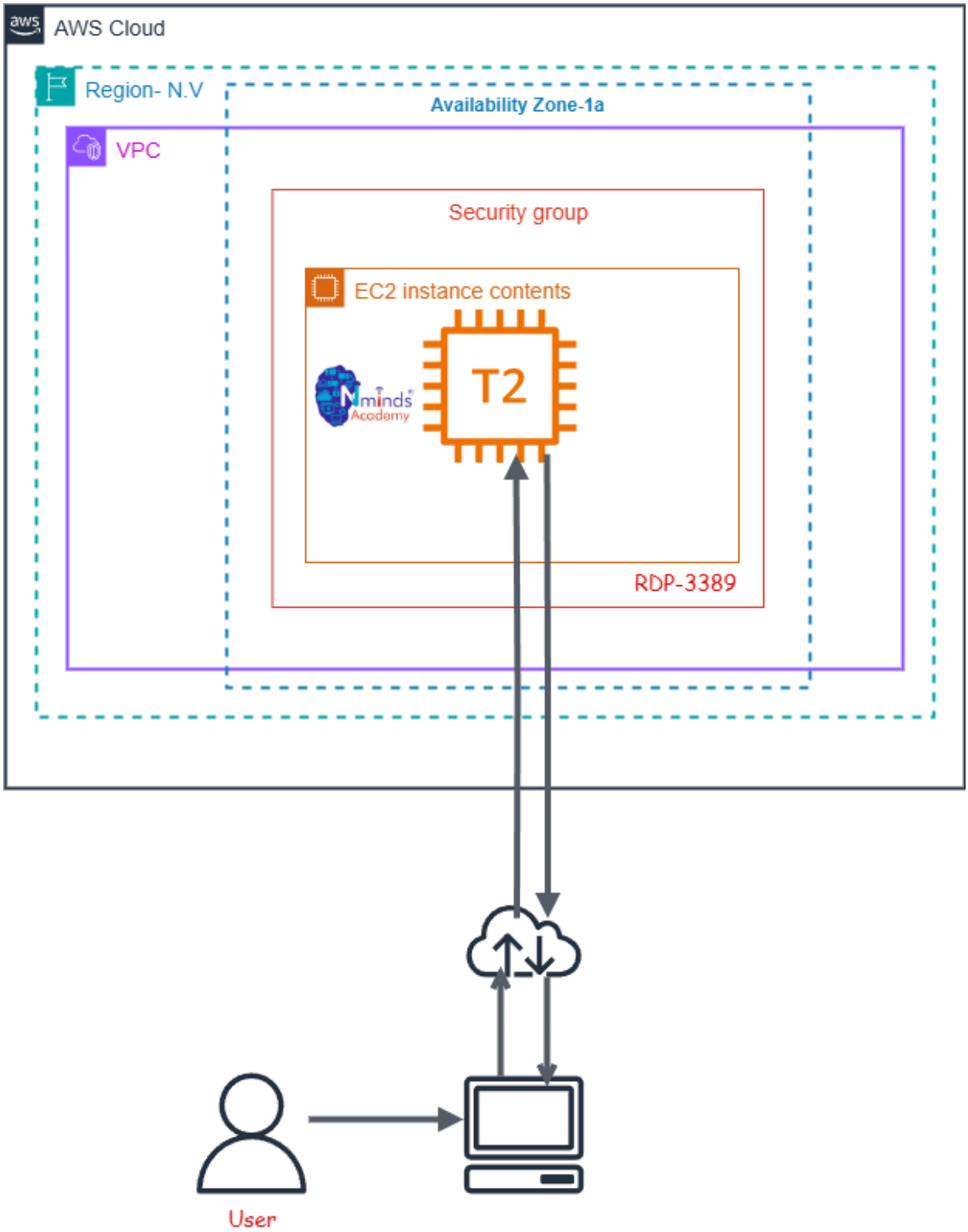
- Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

Used Services in AWS:



Amazon EC2

Topology


AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
RDP-3389
User

Execution Tasks:

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2.Configure IIS Webserver on Windows Server in AWS

Aravindraj.G

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IIS (Internet Information Services) is a web server software created by Microsoft for hosting websites and web applications on Windows Server operating systems. It provides a platform for building, deploying, and managing web-based applications. IIS supports multiple protocols, including HTTP, HTTPS, FTP, and more, and is tightly integrated with the Windows OS.

Common Use Cases of IIS:

1. Hosting Websites: IIS is often used by businesses and developers to host websites and web applications, particularly those built with ASP.NET.
2. Web Applications: For applications that require robust back-end support, IIS provides features like application pooling and security for web services (e.g., SOAP or RESTful APIs).
3. Secure File Sharing: IIS can be configured as an FTP server to securely share files between different locations.
4. Application Hosting in Enterprises: Many enterprise applications built on Microsoft technologies rely on IIS for deployment, such as SharePoint, Exchange Server, or Dynamics CRM.
5. API Hosting: IIS is commonly used to host APIs (e.g., RESTful services) built with .NET or other technologies.

Used Services in AWS

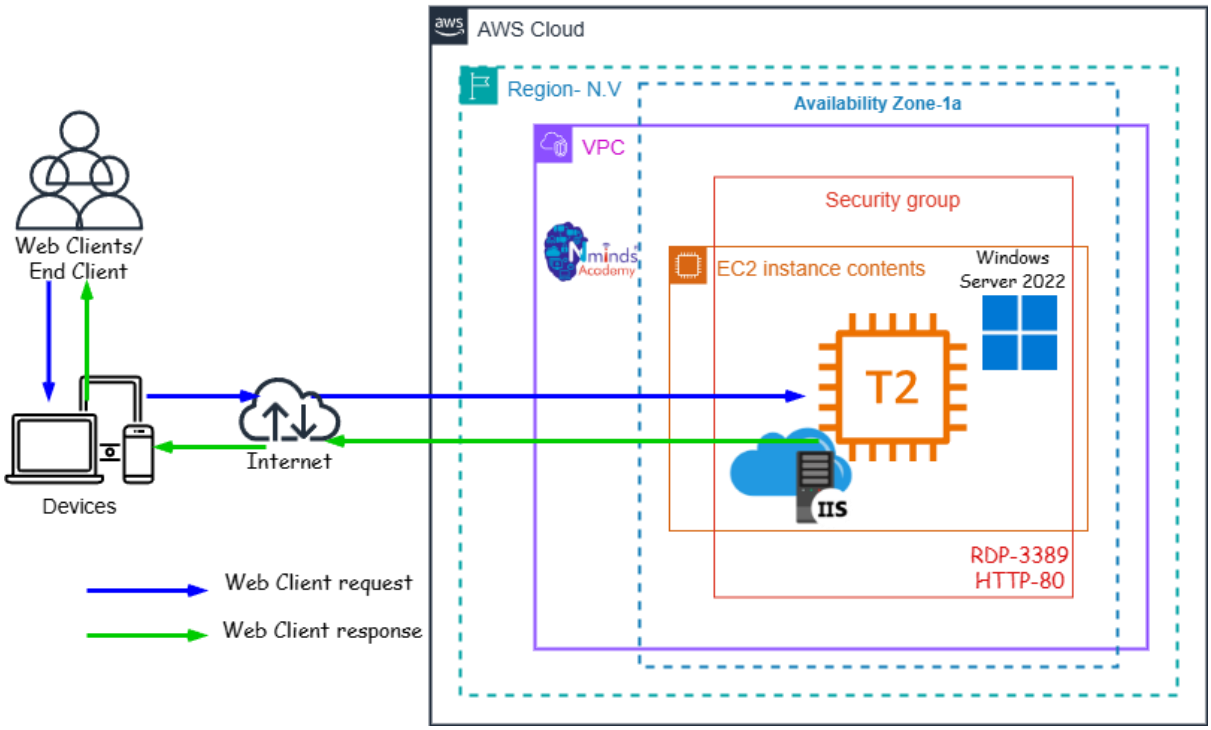
Used Services in Windows Server

Amazon EC2

IIS- Internet Information Services

Topology

How to Configure the IIS web server in Windows Server with AWS EC2



Execution Tasks:

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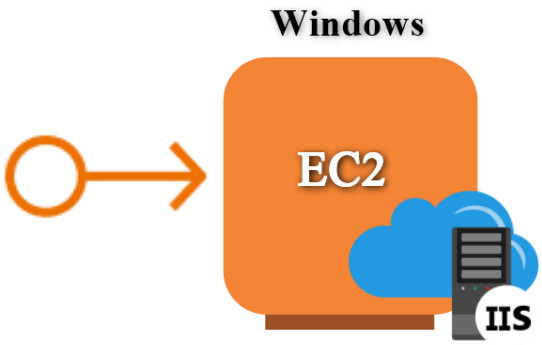
3.Configure Elastic IP Address to Windows Web Server in AWS



Aravindraj.G

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

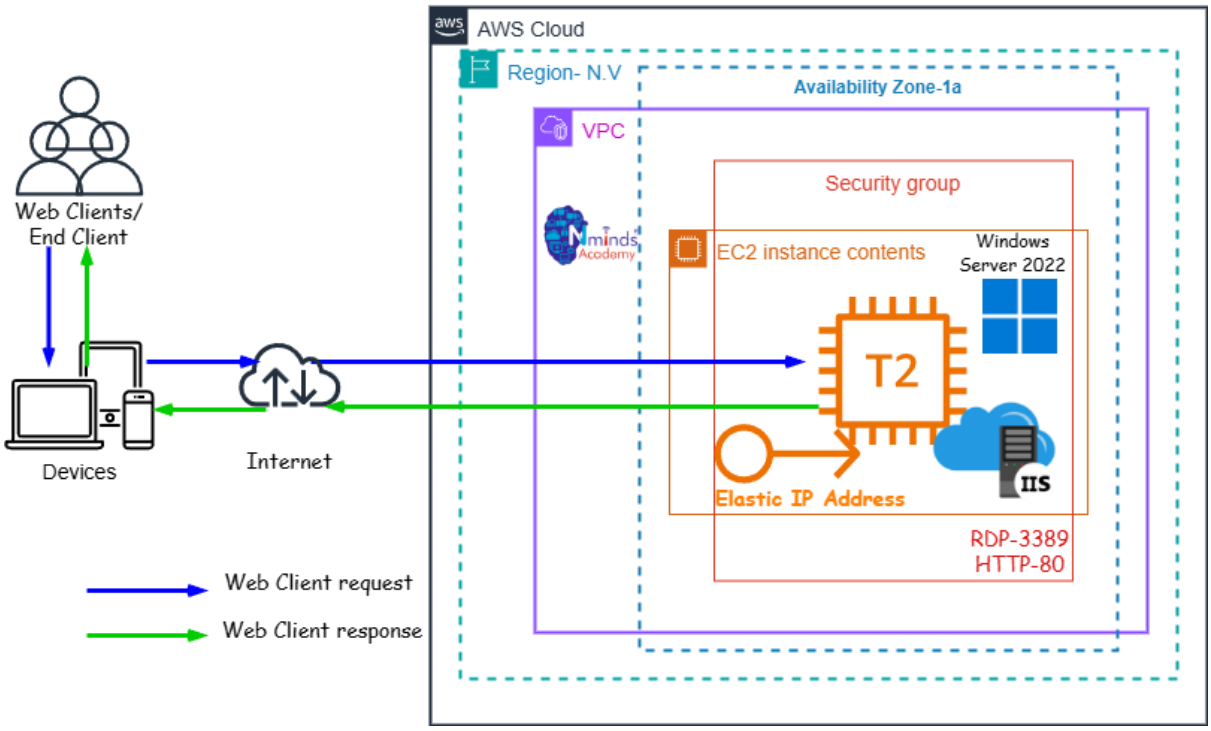
1. Highly Available Applications:
    - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
    - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
    - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology

How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



Used Services in AWS

Used Services in Windows Server

Amazon EC2

IIS- Internet Information Services

Amazon Elastic IP Address

Execution Tasks:

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4.Create an EC2 Linux Instance in AWS

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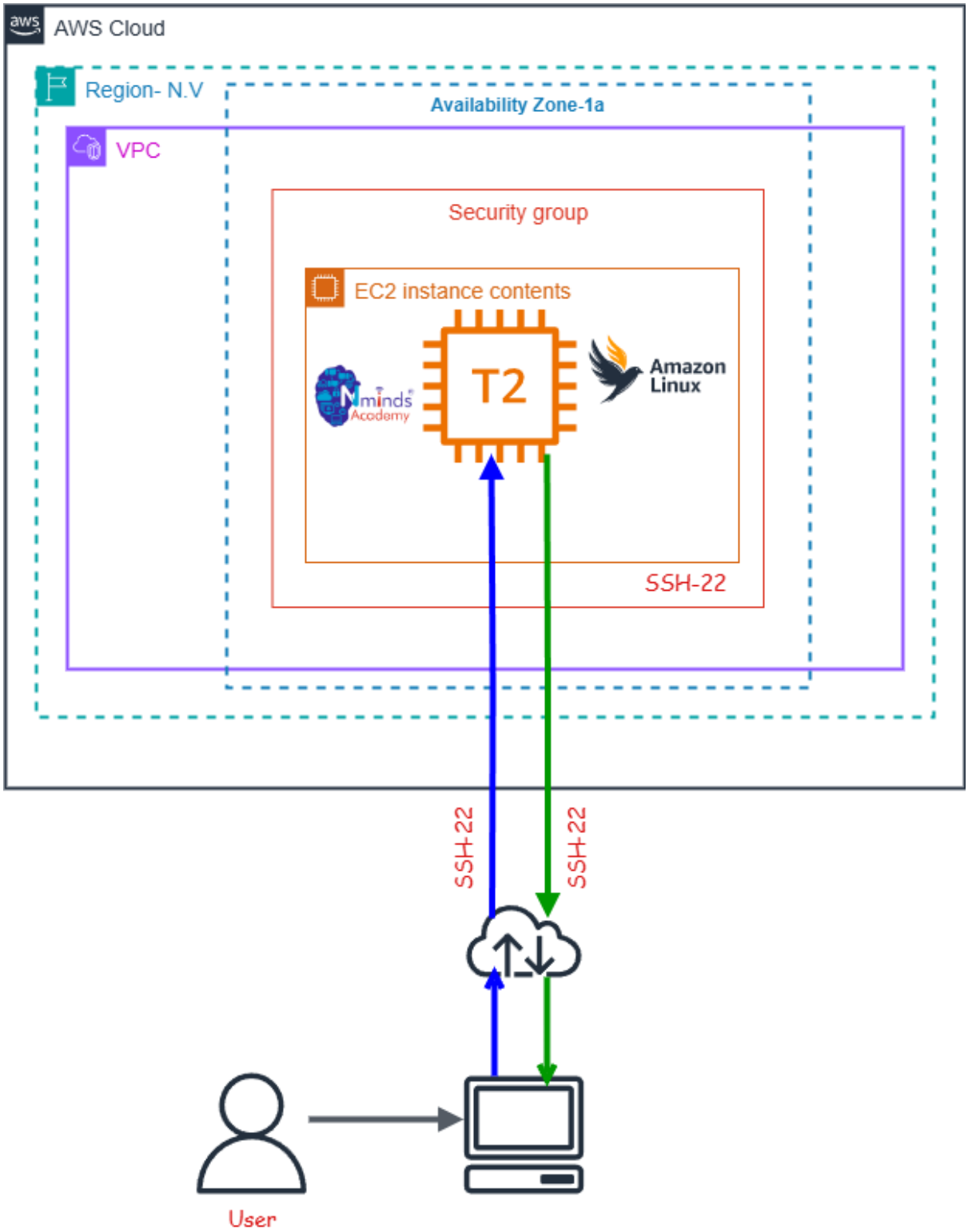
Used Services in AWS

Amazon EC2

Topology



Execution Tasks:

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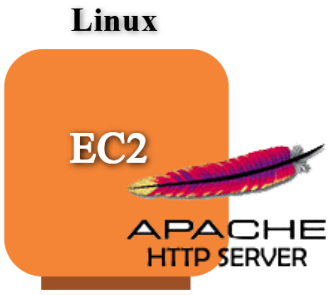
5.Configure Apache Web Server on Linux Instance in AWS



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Used Services in AWS

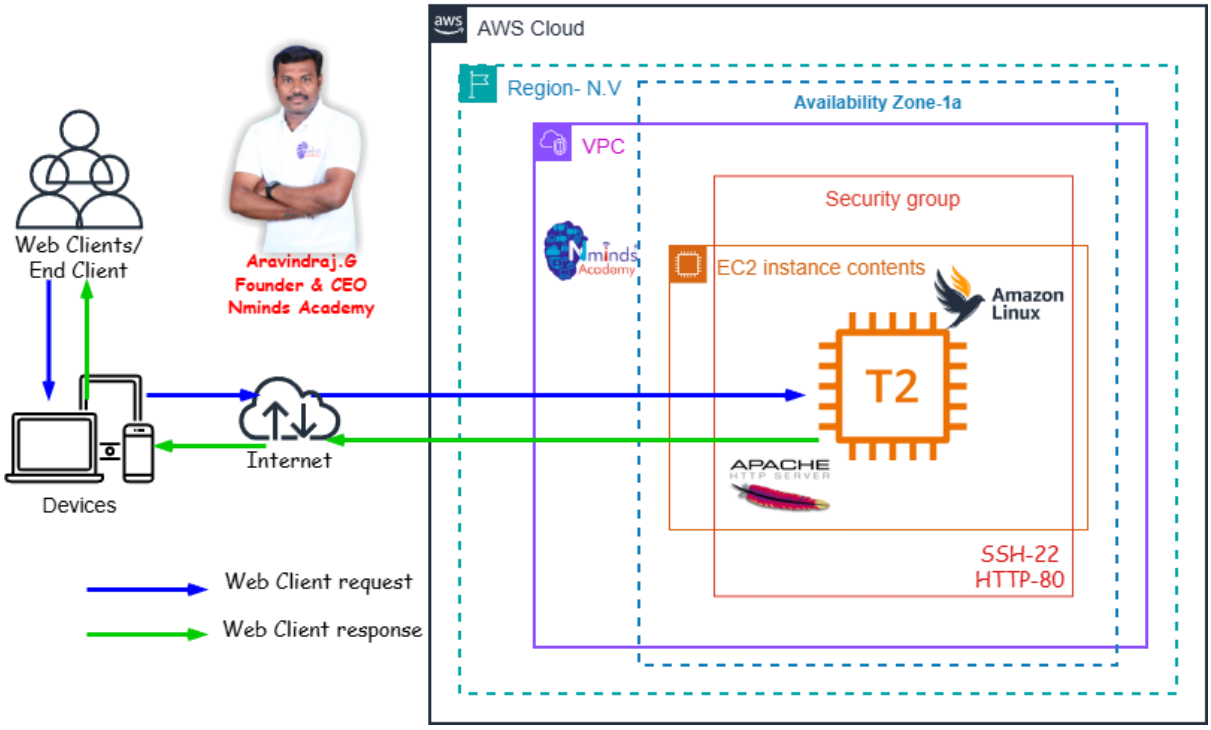
Amazon EC2

Used Services in Linux Server

Apache Webserver

Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:

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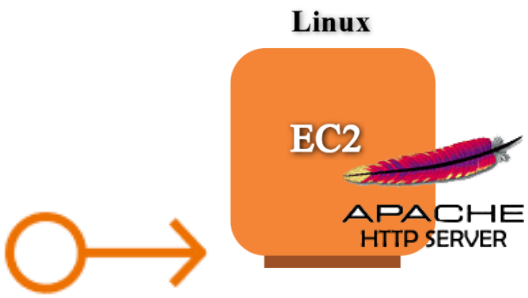
6.Configure Elastic IP Address to Windows Web Server in AWS



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Topology

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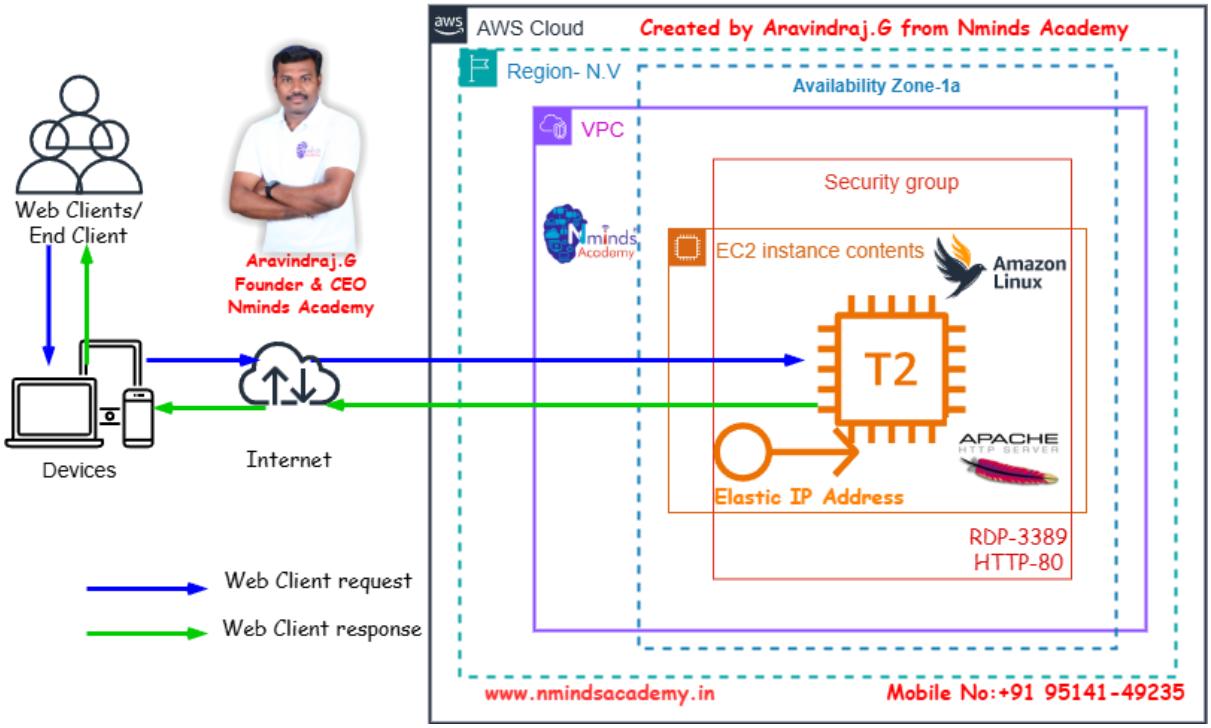

Used Services in AWS

Amazon EC2

Amazon Elastic IP Address

Used Services in Windows Server

Used

Apache Webserver

Execution Tasks:

AWS Solution Architect Training with
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Capstone Projects

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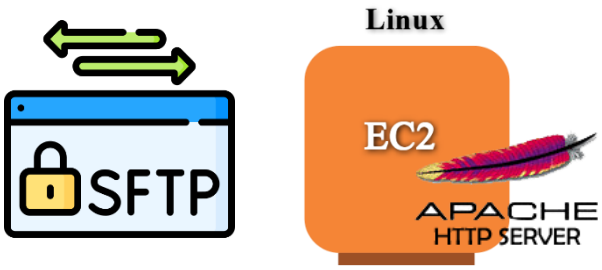
AWS Academy Certified Educator

7.Configure SFTP to Linux Web Server in AWS



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- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

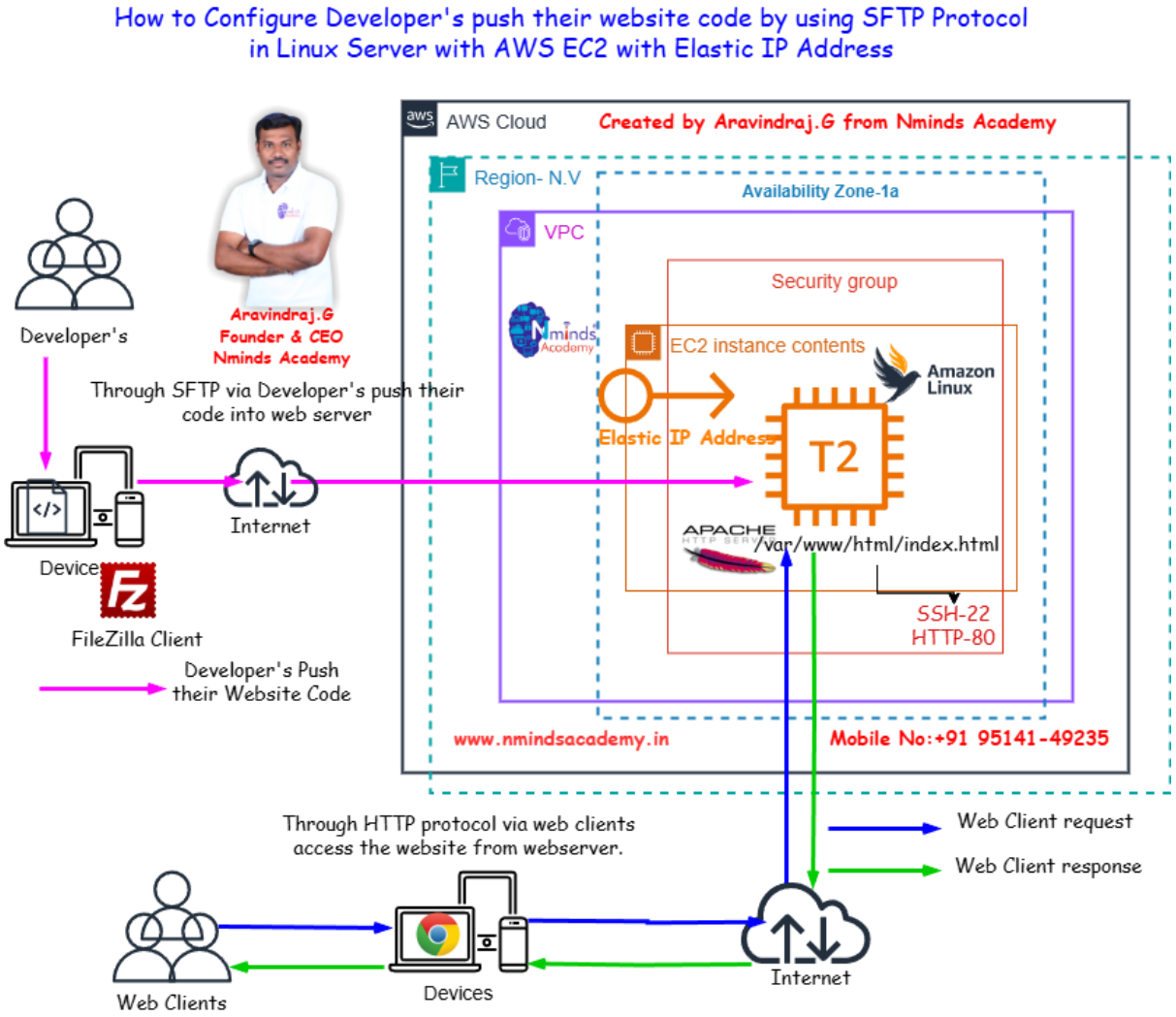
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.

5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology



Used Services in AWS

Services in Windows Server

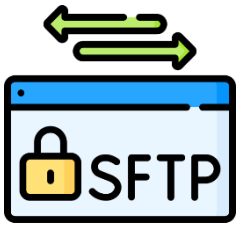
Used

Amazon EC2

APACHE HTTP SERVER

Apache Webserver

Amazon Elastic IP Address



Execution Tasks: